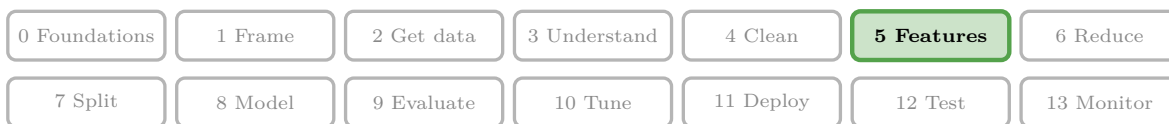


Power Transformer: Box-Cox and Yeo-Johnson Transforms

Contents

1. Overview	2
2. Power transformer in scikit-learn	3
3. Box-Cox transform	3
3.1 The Box-Cox formula	3
3.2 One formula, a whole family of transforms	4
3.3 Finding the best lambda	5
3.4 Only positive values	6
4. Yeo-Johnson transform	6
5. When to use a power transformer	7
6. Power transformer on concrete strength	7
6.1 The data	7
6.2 Baseline: no transform	8
6.3 How normal is each feature?	9
6.4 Box-Cox	11
6.5 Yeo-Johnson	13
6.6 Comparing the lambdas	14
7. Function transformer or power transformer	15
8. Summary	16
9. Sources	16
10. Key terms	16

Where this fits: Pipeline map step 5 (Engineer features). See the [Course map](#).



- **Builds on:** Skewness ([Note 20](#)); Feature transformation ([Note 23](#)); Q-Q plot ([Note 30](#)).
- **Compare with:** Function transformer ([Note 30](#)).

1. Overview

Key point: A power transformer raises each feature to a power λ that it learns from the data, choosing the λ that makes the feature closest to a normal distribution.

A **feature** is an input variable (one column of the data table), the **target** is the output we predict, and an **observation** is one record (one row). Note 30 transformed skewed features with fixed formulas: the log, reciprocal, square and square root. We picked one, tried it, and kept whichever worked best.

The **power transformer** automates that search, like an optician who tries lens after lens until the letters look sharpest. For each feature, the power transformer tries a whole family of power formulas, finds the power λ (lambda) that makes the feature most normal, and applies it.

Figure 1 shows the whole topic. scikit-learn's **PowerTransformer** offers two formulas: the **Box-Cox transform** and the **Yeo-Johnson transform**. The output comes out close to normal, and also standardised.

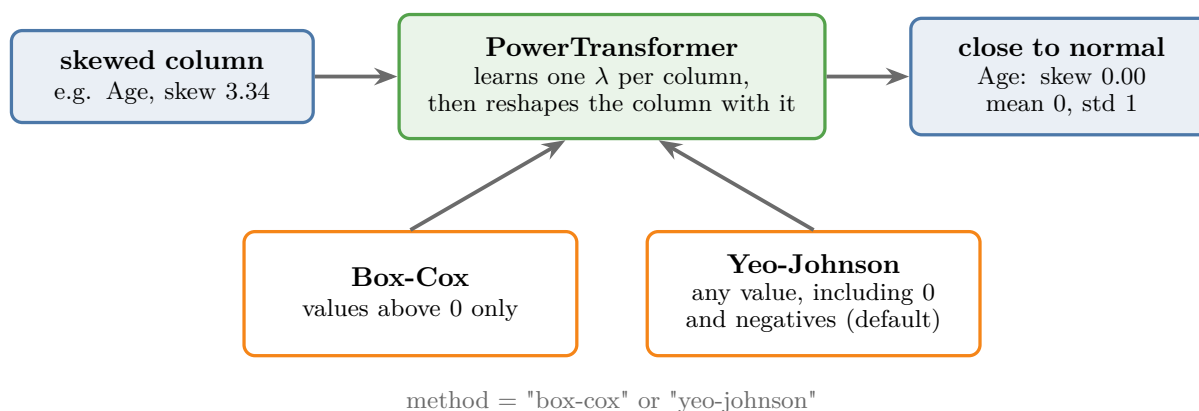


Figure 1: A skewed column goes through PowerTransformer, which learns one lambda per column using Box-Cox or Yeo-Johnson

2. Power transformer in scikit-learn

Key point: `PowerTransformer` is the second of scikit-learn's three mathematical transformers; it holds both the Box-Cox and the Yeo-Johnson transforms.

Note 30 listed scikit-learn's three classes for mathematical transformations:

1. **FunctionTransformer:** applies one formula we choose (Note 30).
2. **PowerTransformer:** applies the Box-Cox or Yeo-Johnson transform, the subject of this Note.
3. **QuantileTransformer:** used much less, and not covered in these Notes.

`PowerTransformer` has one main choice, the parameter **method**. The method is either "box-cox" or "yeo-johnson", and Yeo-Johnson is the default.

3. Box-Cox transform

Key point: Box-Cox raises every value to a power λ ; with the right λ , many skewed features become close to normal.

The Box-Cox transform is named after the two statisticians who published it in 1964, George Box and David Cox (Box and Cox 1964). Box-Cox is one of the most important transforms in practice: with it, a wide range of skewed distributions can be brought close to a normal distribution.

3.1 The Box-Cox formula

Key point: The formula is $(x^\lambda - 1)/\lambda$, and the log when $\lambda = 0$.

The heart of Box-Cox is the power λ . Should a feature become x^2 , x^3 , $x^{1.5}$ or $x^{0.2}$? Box-Cox treats the power as a number to be found from the data.

The Box-Cox transform, step by step:

1. **In words:** raise each value to the power λ , subtract 1, and divide by λ . When λ is exactly 0, that division is impossible, so the log is used instead.
2. **Formula:**

$$x' = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln x & \text{if } \lambda = 0 \end{cases}$$

3. **Example:** for the concrete **Age** feature of Section 6, `PowerTransformer` learns $\lambda = 0.067$. The ages 1, 28 and 365 days become

$$\frac{1^{0.067} - 1}{0.067} = 0, \quad \frac{28^{0.067} - 1}{0.067} = 3.73, \quad \frac{365^{0.067} - 1}{0.067} = 7.24.$$

Before, 365 was 13 times 28; after, it is less than 2 times. The long right tail has been pulled in.

The “subtract 1, divide by λ ” part does not change the shape of the distribution: it only shifts and stretches it. The shape comes from the power λ alone.

Extra: The subtract-and-divide part is there so that the formula flows smoothly into the log as λ gets close to 0. With $\lambda = 0.067$, the value 28 became 3.73, close to $\ln 28 = 3.33$. Without it, $x^{0.001}$ would be almost 1 for every value and the feature would collapse.

3.2 One formula, a whole family of transforms

Key point: The log, square root, reciprocal and square transforms of Note 30 are all Box-Cox with one particular λ .

Each value of λ gives a different transform. Figure 2 draws five of them.

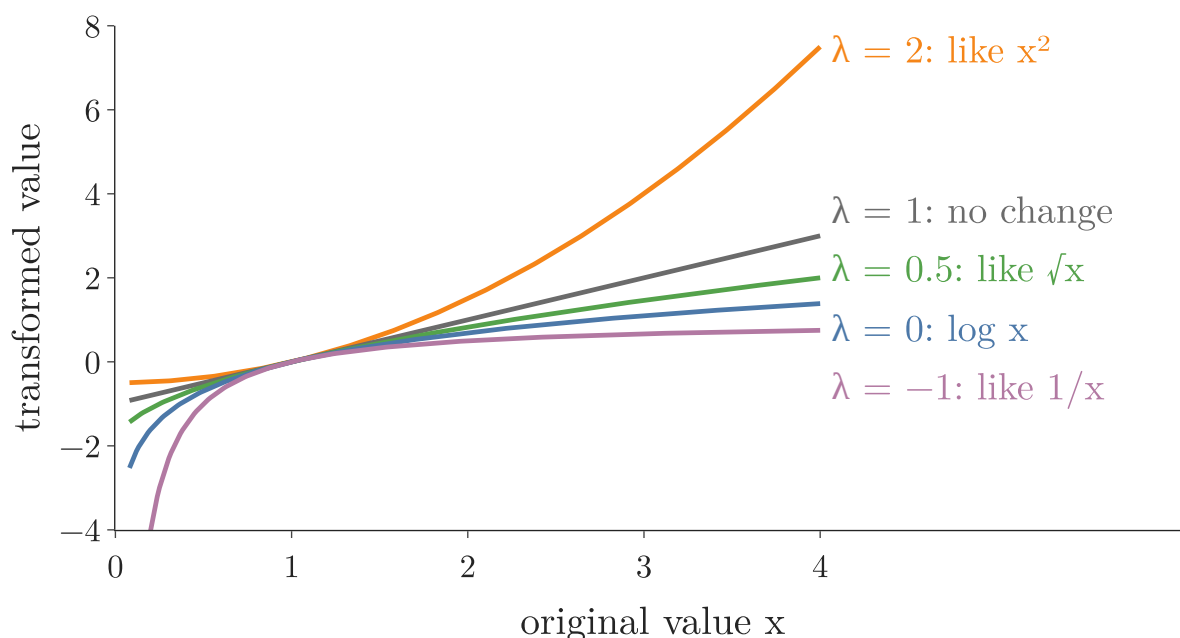


Figure 2: The Box-Cox formula for five values of lambda: each one is a transform from Note 30

λ	Box-Cox gives	Same shape as
2	$(x^2 - 1)/2$	square, for left skew
1	$x - 1$	no change at all
0.5	$2(\sqrt{x} - 1)$	square root
0	$\ln x$	log
-1	$1 - 1/x$	reciprocal

So the transforms of Note 30 are special cases of Box-Cox. The rule is simple:

- λ **below 1**: squashes big values, so it pulls in a right tail. The smaller λ , the stronger the squash.
- λ **above 1**: stretches big values apart, so it pulls in a left tail.
- $\lambda = 1$: leaves the shape as it is.

Unlike Note 30, we no longer have to try each transform by hand. Box-Cox can also land between them, for example on $\lambda = 0.067$, which is a little gentler than the log.

3.3 Finding the best lambda

Key point: Box-Cox tries many values of λ and keeps the one whose result is closest to a normal distribution.

The search goes like this:

1. Try many values of λ , usually between -5 and 5 .
2. Transform the feature with each one.
3. Keep the λ whose result is the best approximation to a normal distribution.

Figure 3 shows the search on the concrete **Age** feature. At $\lambda = 1$ the feature keeps its long right tail (skewness 3.34). Lowering λ pulls the tail in; going too low, to $\lambda = -0.5$, overshoots and creates a left tail instead.

The best value, $\lambda = 0.067$, gives a skewness of almost exactly 0.

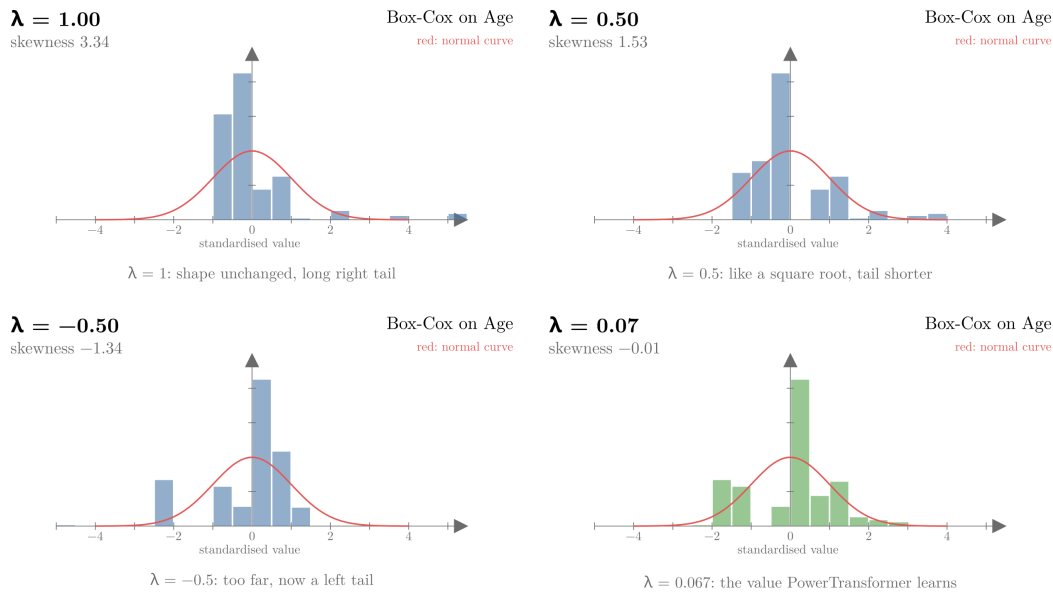


Figure 3: Box-Cox on Age as lambda moves: 1 (unchanged), 0.5, -0.5 (overshoot), and the learned 0.067

Each feature gets its own λ . A dataset with 8 features gets 8 values of λ , one per feature.

Extra: How “closest to normal” is measured. The two standard ways are **maximum likelihood** and Bayesian statistics (Box and Cox 1964). `scikit-learn` uses maximum likelihood (`scikit-learn` docs, **PowerTransformer**): for each λ , it asks “if the transformed values really came from a normal distribution, how likely would we be to see exactly these values?”, and keeps the λ with the highest answer. Maximum likelihood comes back in its own right with logistic regression. A beginner only needs the idea: a computer search picks the most normal-looking λ .

3.4 Only positive values

Key point: Box-Cox works only on values strictly above 0: no zeros, no negatives.

Box-Cox has one important restriction: every value must be strictly positive. A 0 breaks the formula, and so does a negative value.

The reason is in the formula. With $\lambda = 0$ Box-Cox takes $\ln x$, which does not exist for 0 or negative numbers. With other values of λ , a power such as $(-4)^{0.5}$ is not a real number either.

So before using Box-Cox, we check the minimum of every feature.

4. Yeo-Johnson transform

Key point: Yeo-Johnson is a variation of Box-Cox that also works on zeros and negative values.

The Yeo-Johnson transform, published by In-Kwon Yeo and Richard Johnson in 2000 (Yeo and Johnson 2000), is an adjustment of Box-Cox. The main difference is that Yeo-Johnson removes Box-Cox’s restriction: it accepts 0 and negative values.

The Yeo-Johnson transform, step by step:

1. **In words:** for a value of 0 or more, add 1 and then apply Box-Cox. For a negative value, flip its sign, add 1, apply Box-Cox with $2 - \lambda$, and flip the sign back.
2. **Formula:**

$$x' = \begin{cases} \frac{(x+1)^\lambda - 1}{\lambda} & x \geq 0, \lambda \neq 0 \\ \ln(x+1) & x \geq 0, \lambda = 0 \\ -\frac{(1-x)^{2-\lambda} - 1}{2-\lambda} & x < 0, \lambda \neq 2 \\ -\ln(1-x) & x < 0, \lambda = 2 \end{cases}$$

3. **Example:** with $\lambda = 0.5$, the values -3 , 0 and 3 become

$$-\frac{4^{1.5} - 1}{1.5} = -4.67, \quad \frac{1^{0.5} - 1}{0.5} = 0, \quad \frac{4^{0.5} - 1}{0.5} = 2.$$

Zero stays at 0, and the negative value is handled without any error.

For data with no negative values, only the first two lines apply. Then Yeo-Johnson is Box-Cox applied to $x + 1$, just as $\log(1 + x)$ in Note 30 was the log applied to $x + 1$.

Extra: Why the negative side uses $2 - \lambda$. A λ below 1 squashes the right tail; for the transform to treat both tails alike, the left side must squash in the mirror-image way. Using $2 - \lambda$ on the flipped values does exactly that, so the curve stays smooth through 0.

5. When to use a power transformer

Key point: When a feature is not normal and the algorithm prefers normal data, apply both transforms and keep the better one.

Note 30 explained which algorithms care about the shape of the data: linear regression and logistic regression do; decision trees and random forests do not. The same rule decides when to use a power transformer.

The recipe:

1. Check each feature's distribution (density plot, skewness, Q-Q plot from Note 30).
2. If features are not normal and the algorithm is a linear one, apply a power transformer.
3. Try both methods, Box-Cox and Yeo-Johnson, just like trying settings of a model, and keep the one that scores better.

In real-world data, features are rarely normal to begin with, so this check is worth doing every time.

6. Power transformer on concrete strength

Key point: On the concrete data, linear regression's R^2 rose from 0.63 to 0.80 with Box-Cox and 0.82 with Yeo-Johnson.

6.1 The data

Key point: 1030 concrete mixes, 8 numerical features and the target **Strength**: a regression problem.

The dataset describes 1030 batches of concrete; each batch is one observation. Several of its features are clearly not normal, which makes it a good test for a power transformer.

Column	Meaning
Cement, Blast Furnace Slag, Fly Ash, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate	amount of each ingredient in the mix (kg per cubic metre)
Age	how many days the concrete has set, from 1 to 365

Column	Meaning
Strength	how much pressure the concrete withstands (the target)

Strength is a number, so this is a regression problem, and the model is linear regression.

Python: Loading and checking the data.

```
import pandas as pd

df = pd.read_csv("data/concrete_data.csv")
df.shape           # (1030, 9)
df.isnull().sum()  # all 0: no missing values
df.describe()      # look at the "min" row
```

There are no missing values, which matters: transforms cannot handle them. The `min` row of `describe()` shows the problem for Box-Cox:

Column	Minimum	Zeros
Blast Furnace Slag	0	471
Fly Ash	0	566
Superplasticizer	0	379

No feature is negative, but three contain many zeros.

As always, we split first: 80% for training (824 rows) and 20% for testing (206 rows), with `random_state=42`.

Extra: The data is the “Concrete Compressive Strength” dataset by I-Cheng Yeh (Yeh 1998), from the UCI Machine Learning Repository. The full file is small (about 60 KB), so it is kept whole in `data/`.

6.2 Baseline: no transform

Key point: With no transform, linear regression scores R^2 0.63 on the test set and 0.46 with cross-validation.

To measure a regression model, we use the **R^2 score**: how much of the variation in the target the model explains. A score of 1 is perfect; a score of 0 is no better than always predicting the average strength. R^2 has its own Note later; here we only compare its values.

Check	R^2
Test set	0.628
5-fold cross-validation	0.461

Cross-validation, from Note 30, averages the score over 5 different splits. Cross-validation gives a lower and more honest number here, 0.46.

Python: The baseline.

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from sklearn.model_selection import cross_val_score

lr = LinearRegression()
lr.fit(X_train, y_train)
r2_score(y_test, lr.predict(X_test))          # 0.628
cross_val_score(LinearRegression(), X, y,
                 scoring="r2", cv=5).mean()    # 0.461
```

Extra: The features are not scaled here. Leaving them unscaled is fine: plain linear regression gives exactly the same predictions with or without scaling, because it simply adjusts its coefficients to the units. The Notebook checks this (R^2 0.628 either way). Scaling matters for models such as KNN, or regression trained step by step with gradient descent.

6.3 How normal is each feature?

Key point: Age is the worst feature (skewness 3.34); Blast Furnace Slag, Fly Ash and Superplasticizer are piled up at 0; the two aggregates are already close to normal.

Figure 4 shows each training feature as a histogram (skewness in brackets) with its Q-Q plot below.

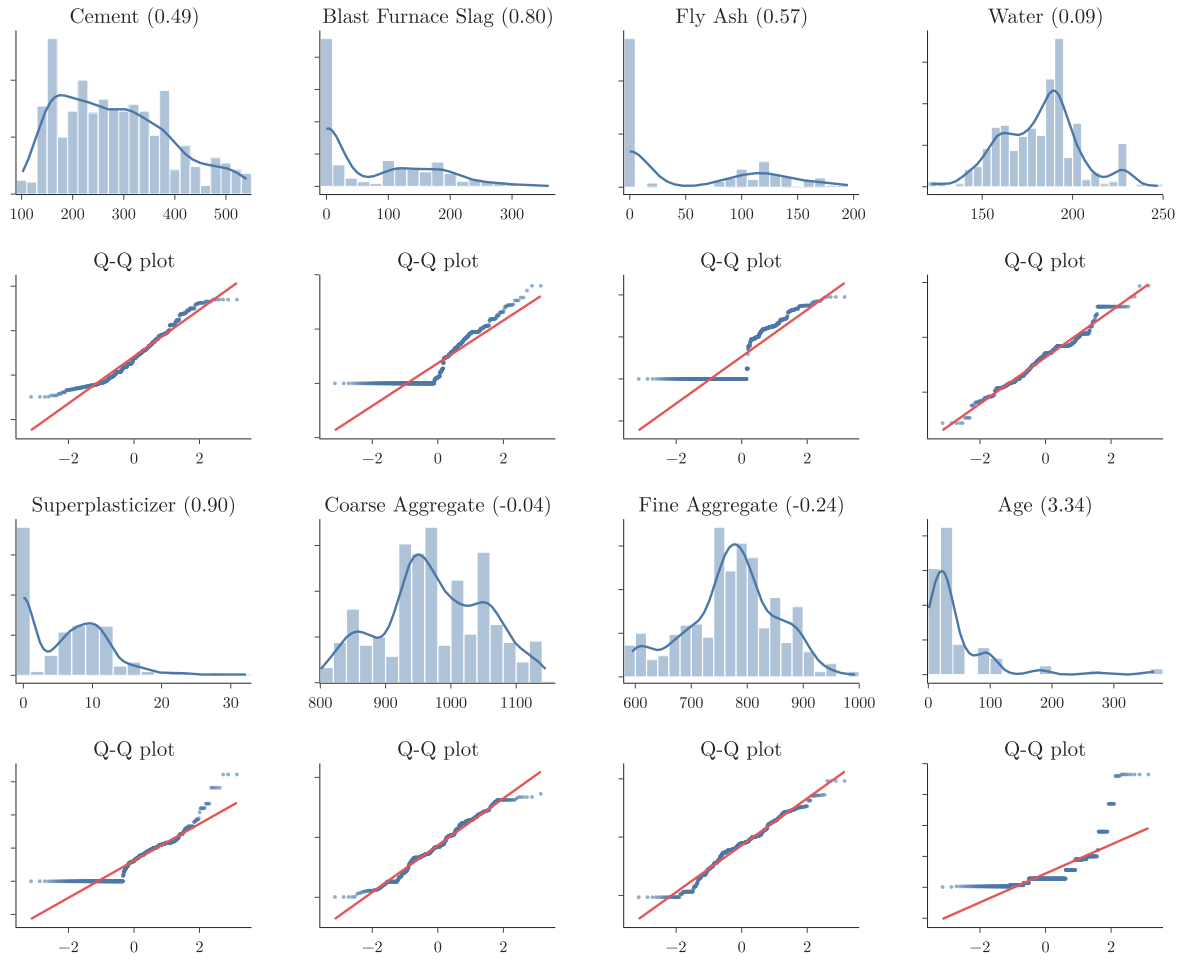


Figure 4: The 8 raw input columns of the training set: histogram (skewness in brackets) and Q-Q plot

- **Cement:** slightly right-skewed, but fairly close to normal.
- **Blast Furnace Slag:** right-skewed, with a huge pile of zeros: the flat run of points at the bottom of its Q-Q plot.
- **Fly Ash:** **bimodal**, meaning it has two peaks: one at 0 and one around 120.
- **Water:** reasonably close to normal.
- **Superplasticizer:** right-skewed and bimodal, again with a pile of zeros.
- **Coarse Aggregate:** the closest to normal; its points follow the line.
- **Fine Aggregate:** also close to normal.
- **Age:** the worst. Most mixes are tested young (28 days or less), and a few after a whole year, so there is a long right tail.

6.4 Box-Cox

Key point: Adding a tiny 0.000001 to every value lets Box-Cox run despite the zeros; it then lifted R^2 to 0.80 on the test set and 0.67 with cross-validation.

Box-Cox cannot run on the three features with zeros. A common workaround is to add a tiny number to every value: a 0 becomes 0.000001, which is above 0, and every other value hardly changes.

Python: Box-Cox with PowerTransformer.

```
from sklearn.preprocessing import PowerTransformer
```

```
pt = PowerTransformer(method="box-cox")
```

```
X_train_transformed = pt.fit_transform(X_train + 0.000001)
```

```
X_test_transformed = pt.transform(X_test + 0.000001)
```

```
pt.lambdas_          # one lambda per column
```

fit learns one λ per feature from the training set; transform applies those same λ values to both sets. The learned values are stored in `lambdas_`.

The attribute `lambdas_` holds the λ learned for each feature, in column order (listed in Section 6.6). For `Cement` it is 0.177, so every cement value x becomes $(x^{0.177} - 1)/0.177$: a value of 540 becomes 11.6, and 102 becomes 7.2.

Training the same linear regression on the transformed features:

Check	No transform	Box-Cox
Test set R^2	0.628	0.805
Cross-validated R^2	0.461	0.666

The cross-validated R^2 rose by about 0.2, a very good improvement for one line of preprocessing.

Figure 5 shows the feature that changed most, `Age`. The long right tail is gone, the skewness drops from 3.34 to almost 0, and the Q-Q plot points now follow the line.

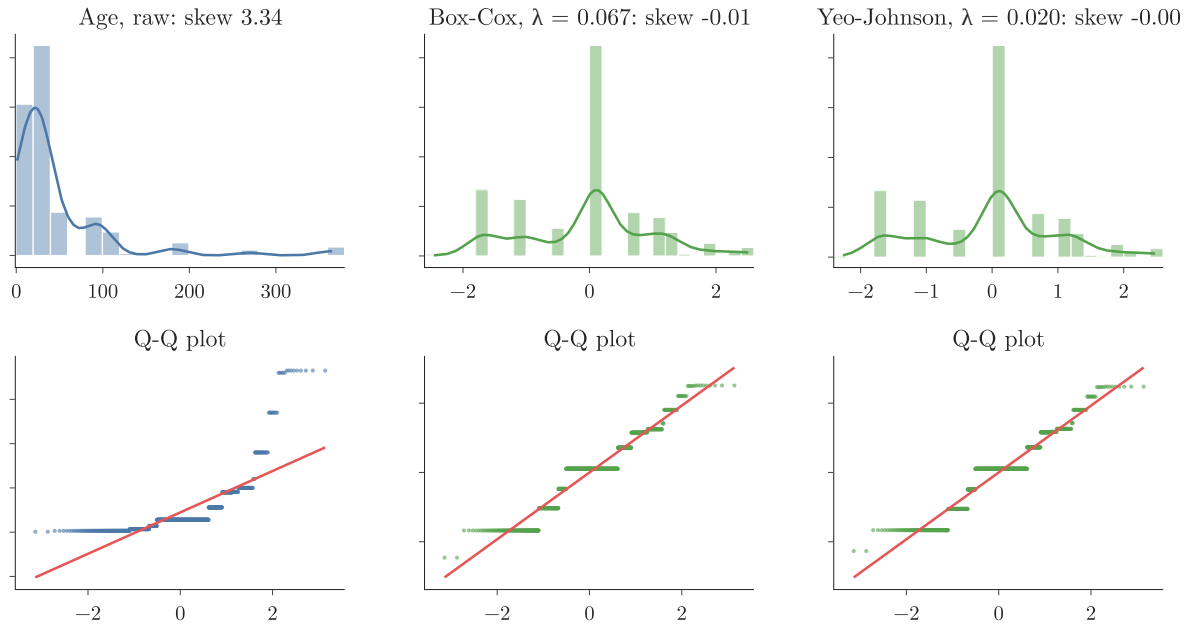


Figure 5: Age before (blue) and after Box-Cox and Yeo-Johnson (green): histogram and Q-Q plot

The spikes that remain are because **Age** takes only 14 different values (1, 3, 7, 14, 28 days and so on). A transform moves each value, but it cannot split one value into several.

Figure 6 shows all 8 features after Box-Cox.

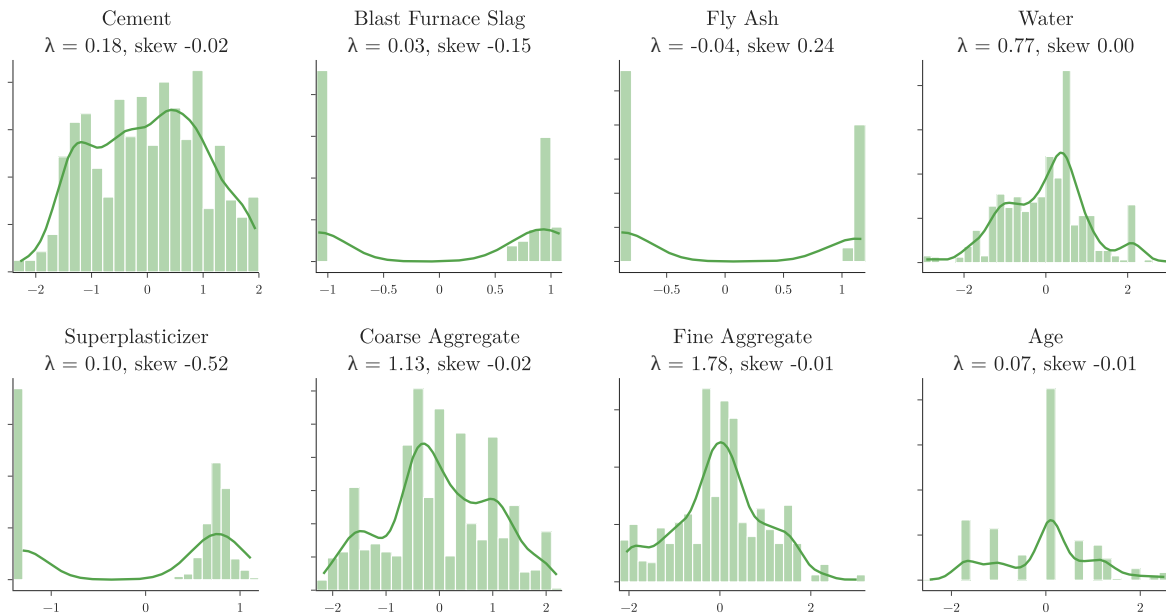


Figure 6: All 8 training columns after Box-Cox, with each learned lambda and skewness

- **Cement** and **Superplasticizer**: improved, closer to a bell shape.
- **Blast Furnace Slag** and **Fly Ash**: not made normal. The two separate groups (zero and non-zero) are still two groups.
- **Water**, **Coarse Aggregate** and **Fine Aggregate**: almost unchanged; they were close to normal already.
- **Age**: the biggest improvement.

Extra: The tiny-number trick has a side effect. With λ near 0, Box-Cox behaves like a log, and $\ln(0.000001) = -13.8$, far below the log of any real amount (at least $\ln 11 = 2.4$ for slag). So all the zeros are thrown far to the left, as a separate block. The thrown zeros are why Blast Furnace Slag and Fly Ash in Figure 6 look like two bars at opposite ends. Yeo-Johnson does not need the trick, and handles zeros more gently.

6.5 Yeo-Johnson

Key point: Yeo-Johnson needs no workaround for zeros and scored slightly better: R^2 0.82 on the test set and 0.68 with cross-validation.

Yeo-Johnson is the default method, so `PowerTransformer()` with no arguments uses it. Yeo-Johnson accepts the zeros as they are.

Python: Yeo-Johnson.

```
pt1 = PowerTransformer()    # method="yeo-johnson"
X_train_transformed2 = pt1.fit_transform(X_train)
X_test_transformed2 = pt1.transform(X_test)
```

Check	No transform	Box-Cox	Yeo-Johnson
Test set R^2	0.628	0.805	0.816
Cross-validated R^2	0.461	0.666	0.683

So the score went from 0.46 with no transform, to 0.67 with Box-Cox, to 0.68 with Yeo-Johnson.

Figure 7 shows all 8 features after Yeo-Johnson.

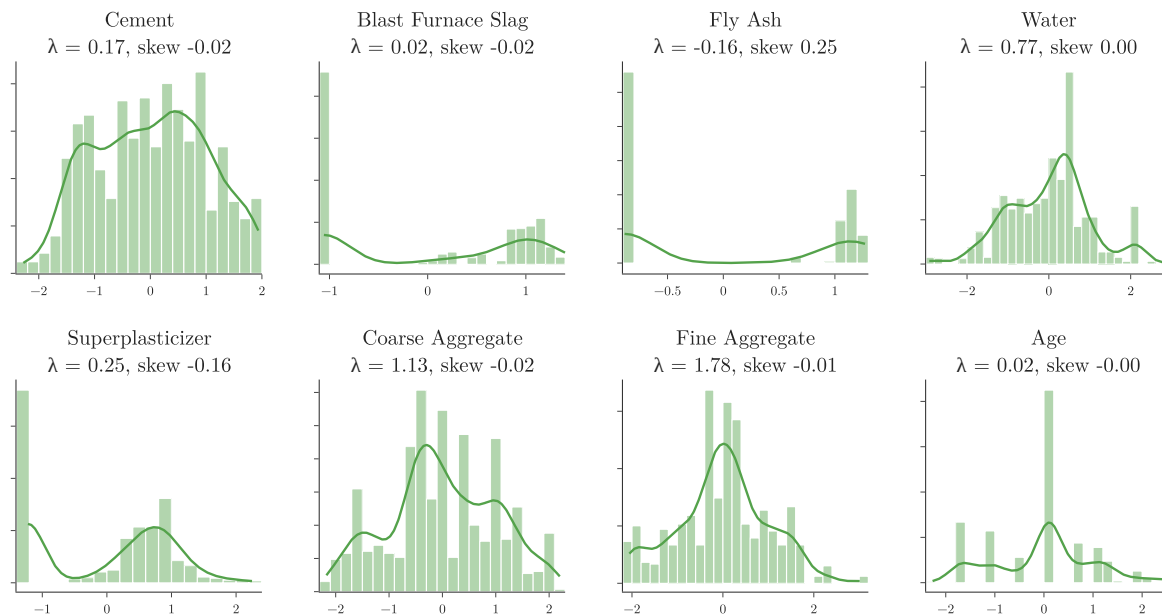


Figure 7: All 8 training columns after Yeo-Johnson, with each learned lambda and skewness

The result looks much like Box-Cox in Figure 6. The visible differences are in the features with zeros: Superplasticizer and Blast Furnace Slag are spread out more smoothly, because Yeo-Johnson maps a 0 to 0 instead of throwing it far away. There are no negative values in this data, so Yeo-Johnson’s main advantage is not even used.

Extra: `PowerTransformer` also standardises its output by default (parameter `standardize=True`): after the power transform, every feature gets mean 0 and standard deviation 1, as with the `StandardScaler` of Note 24 (scikit-learn docs, `PowerTransformer`). Standardising is why the axes in Figures 6 and 7 run from about -2 to 2 .

6.6 Comparing the lambdas

Key point: The two methods learn almost the same λ for most features; they differ most on the features with zeros.

Both transformers fitted on the same training set:

Column	Box-Cox λ	Yeo-Johnson λ	Skewness before	After Yeo-Johnson
Cement	0.177	0.174	0.49	−0.02
Blast Furnace Slag	0.025	0.016	0.80	−0.02
Fly Ash	−0.039	−0.161	0.57	0.25
Water	0.773	0.771	0.09	0.00

Column	Box-Cox λ	Yeo-Johnson λ	Skewness before	After Yeo-Johnson
Superplasticizer	0.099	0.254	0.90	-0.16
Coarse Aggregate	1.130	1.130	-0.04	-0.02
Fine Aggregate	1.782	1.783	-0.24	-0.01
Age	0.067	0.020	3.34	0.00

The table follows the rule of Section 3.2:

- **Right-skewed features** (Cement, Slag, Fly Ash, Superplasticizer, Age) get λ well below 1, close to a log.
- **Water and Coarse Aggregate**, already near normal, get λ near 1: little change.
- **Fine Aggregate**, slightly left-skewed, gets $\lambda = 1.78$, close to a square.

The biggest gaps between the two methods are on Fly Ash and Superplasticizer, two of the features with zeros: the tiny-number trick changes what Box-Cox sees there.

Extra: The cross-validated scores above fit the power transformer on all of $\mathbf{X}$ before cross-validating, so each test fold has already influenced the λ values: a small case of data leakage. Putting the transformer inside a pipeline (Note 29) refits it on each training fold. The honest scores are 0.660 for Box-Cox and 0.674 for Yeo-Johnson, almost the same, so the conclusion holds.

```
from sklearn.pipeline import make_pipeline

pipe = make_pipeline(PowerTransformer(),
                    LinearRegression())
cross_val_score(pipe, X, y, scoring="r2",
                cv=5).mean()    # 0.674
```

7. Function transformer or power transformer

Key point: The power transformer usually beats a hand-picked function such as the log, but the only way to know is to try both.

The two transformers of Notes 30 and 31 do the same job in different ways:

- **FunctionTransformer:** we pick the formula (log, square root and so on) and apply it.
- **PowerTransformer:** searches a whole family of formulas and picks the best power for each feature.

Because the power transformer chooses from many formulas, including the log and square root themselves, it usually performs better. Still, try both and keep whichever gives the better score on the project at hand.

8. Summary

	Function transformer	Box-Cox	Yeo-Johnson
Formula	one we choose	$(x^\lambda - 1)/\lambda$, or $\ln x$	Box-Cox on $x + 1$, mirrored for negatives
Power λ	none	learned per column	learned per column
Values allowed	depends on the formula	above 0 only	any
In scikit-learn	FunctionTransformer	PowerTransformer, method “box-cox”	PowerTransformer, the default method
Concrete data, cross-validated R^2	(not tried)	0.666	0.683

- A power transformer learns one power λ for each feature and uses it to bring the feature close to a normal distribution.
- Box-Cox: $(x^\lambda - 1)/\lambda$, or $\ln x$ when $\lambda = 0$. The log, square root, reciprocal and square are special cases.
- The best λ is found by a search (maximum likelihood) and stored in `lambda_`.
- λ below 1 pulls in a right tail; above 1 pulls in a left tail; 1 leaves the shape alone.
- Box-Cox needs values above 0; Yeo-Johnson works on any value and is the default.
- `PowerTransformer` also standardises its output to mean 0 and standard deviation 1.
- On the concrete data, linear regression’s cross-validated R^2 rose from 0.46 to 0.67 (Box-Cox) and 0.68 (Yeo-Johnson). Age gained most: skewness 3.34 to 0.
- A transform cannot merge two separate groups, such as a pile of zeros and the rest, into one bell.

9. Sources

- Box, G. E. P. and Cox, D. R. (1964). An Analysis of Transformations. *Journal of the Royal Statistical Society, Series B* 26(2), 211-252.
- scikit-learn documentation. `sklearn.preprocessing.PowerTransformer`. scikit-learn.org.
- Yeh, I-C. (1998). Modeling of Strength of High-Performance Concrete Using Artificial Neural Networks. *Cement and Concrete Research* 28(12), 1797-1808. Data: UCI Machine Learning Repository, Concrete Compressive Strength.
- Yeo, I-K. and Johnson, R. (2000). A New Family of Power Transformations to Improve Normality or Symmetry. *Biometrika* 87(4), 954-959.

10. Key terms

Term	Meaning
Feature	An input variable, one column of the data table
Target	The output we predict
Observation	One record, one row of the data table
Power transformer	A transform that raises each feature to a learned power λ to make it close to normal
PowerTransformer	scikit-learn's class for the Box-Cox and Yeo-Johnson transforms
Lambda (λ)	The power used by a power transform, learned separately for each feature
Box-Cox transform	$(x^\lambda - 1)/\lambda$, or $\ln x$ when $\lambda = 0$; works only on values above 0
Yeo-Johnson transform	A variation of Box-Cox that also works on zero and negative values; scikit-learn's default
method	The PowerTransformer parameter that picks "box-cox" or "yeo-johnson"
lambdas__	The PowerTransformer attribute holding the learned λ of each feature
standardize	The PowerTransformer parameter (on by default) that rescales the output to mean 0 and standard deviation 1
Maximum likelihood	Choosing the parameter value under which the observed data is most likely; used to find λ
Bimodal	A distribution with two peaks
R ² score	How much of the variation in a regression target the model explains: 1 is perfect, 0 is no better than the average